

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 216 — Past Questions & Materials
by Renance

Course: MEE 216
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200

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Paper 1 of 2 — MEE 216 PAST QUESTIONS

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MECHANICAL ENGINEERING

First Semester Examination 2017/2018 Session

COURSE CODE AND TITLE: MEE 216 - BASIC THERMODYNAMICS

INSTRUCTION: Answer question one (1) is compulsory and any other three questions. TIME

ALLOTTED: 2.5Hrs

Question 1 (25 marks)

(a) Define any five of following terms: (i) path function, (ii) superheated. vapour, (iii) critical point, (iv)

thermodynamic equilibrium, (v) perpetual motion machine of the first kind (PPM1), (vi) weak mixture, (vi)

refrigeration capacity, (vii) throttling process and (viii) coefficient of performance

(b) Highlight reason why the Carnot cycle is considered impracticable in real life.

(c) Itemize the basic laws upon which the study of the thermodynamics is based.

(d) A closed system changes from state 1 to state 2 while 40 kJ of heat is added and 60 kJ of work is done

by the system. As the system is returned to state 1, 35 kJ of work is done on it. What is the heat transfer during process 2-1?

(e) (i) Compare and contrast the difference between Carnot, Otto, diesel and actual cycles or (ii) distinguish

between intensive and extensive properties.

Question 2 (15 marks)

(a) Compare and contrast the difference between isothermal process and adiabatic process.

(b) With the aid of a labeled diagram, explain Carnot cycle.

(c) A gas expands reversibly according to the relation $PV^n = C$. The initial volume and pressure of the gas are

0.0283 m³ and 13.8 bar respectively. The final pressure and volume are 1.035 bar and 0.2073 m³.

Calculate the polytropic index of expansion and determine the work done during the process.

~ Question 3 " (15 marks)

(a) Distinguish between a compressor and a pump or a nozzle and a diffuser.

(b) Distinguish between a homogeneous system and a heterogeneous system.

(c) A heat engine rejects 600 kJ of heat energy as it does 400 KJ of work. What is the thermal efficiency of the heat engine.

"Question 4 (15 marks)

(a) Differentiate between closed system, open system and an isolated system.

(b) State three similarities and differences between work and heat.

(c) 0.92 kg of a certain perfect gas at a temperature of 150 °C and pressure of 6.25 bar is constrained in a piston-cylinder assembly and occupies a volume of 0.185 m³. The gas is allowed to expand at constant pressure until the volume is 0.85 m³. Calculate (i) the molecular mass, (ii) the final temperature, and (iii) the work done by the gas. (Take $R = 8.314 \text{ kJ/Kmol/K}$)

Question 5 (15 marks)

- (a) What is the usefulness of the ideal Rankine cycle or why is it introduced?
- (b) Define the following; (i) pure substance, (ii) triple point, (iii) sub-cooled liquid and quality liquid
- (c) (1) Draw the following process on the PV-plane and derive an expression for the work done on (i) polytrophic process, (ii) isobaric process and (iii) isothermal process OR (2) Distinguish between macroscopic and microscopic approaches. . cz

-Question 6 (15 marks) Wwe W: Fe preleg-con~

- (a) How is evaporation different from boiling?
- (b) Comment on the statement "Energy is always conserved, but its quality is always neglected."
- (c) A fluid undergoes a reversible adiabatic compression from 4 bar, 0.3 m³ to 0.08 m³, according to the law, $p v^{1.25} = \text{constant}$. Determine (i) change in enthalpy; (ii) change in internal energy; (iii) heat transfer, and (iv) work transfer -

FEDERAL UNIVERSITY OF PENINSULAR RESOURCES, EFFURUN

DEPARTMENT OF AEROSPACE ENGINEERING

First Semester Examination @0D6/2017 Academic Session

COURSE CODE AND TITLE: MBE 216 - BASIC-THERMODYNAMICS

DATE: APRIL 1ST, 2017_ TIME ALLOWED: 2 HOURS

INSTRUCTIONS: ANSWER #IN FOUR (4) QUESTIONS.

QUESTION 1

- (a) Mention any four (4) terms/statements essential for inclusion in a complete definition of Thermodynamics. [2 Marks]
- (b) Highlight one major difference between Classical Thermodynamics and Statistical Thermodynamics. [2 Marks]
- (c) Is it true the assertion: "all closed systems are adiabatic"? Justify your answer. [2 Marks]
- (d) How many thermodynamic states would be contained in a thermodynamic cycle with five (5) distinct processes? Illustrate the cycle on a property (P-V) diagram, labeling clearly the states and processes. [4 Marks]
- (e) Could the mixture of liquid air and gaseous air be regarded as pure substance? Justify your answer. [2 Marks]

(f) A temperature scale of certain thermometer is given by the relation $t = x \ln p + y$, where x and y are constants

and p is the thermometric property of the fluid in the thermometer. If at the ice point and steam point the

thermometric properties are found to be 2.5 and 8.0 respectively, what will be the temperature corresponding to the thermometric property of 4.0 on Celsius scale? [5.5 Marks]

QUESTION 2

(a) Mention the basic forms in which energy moves through the boundaries of thermodynamic systems. [2

Marks]

(b) If a Linear Fresnel solar collector system releases 250 kW of heat, out of the total 280 kW it absorbs from

the Sun, as part of the processes required to produce 40 kW of electricity. What are the values of thermal

efficiency and output from the solar collector system? [2 Marks]

(c) State the mathematical expression of the First law of Thermodynamics, as it applies to a closed system.

[1 Mark]

(d) Consider an ideal thermal power plant, based on Organic Rankine Cycle (ORC), which operates with silicon oil,

If 200 kW of heat is used to evaporate the silicon oil inside the ORC, and the turbine component of the ORC

produces 160 kW of mechanical work; determine the value of internal energy capacity of the ORC system,

assuming it is closed. Identify the working substance of the system. [5 Marks]

(e) A diesel engine piston which has an area of $45 \times 10^{-3} \text{ m}^2$ moves 5 cm during part of suction stroke, 300×10^{-6}

m^3 of fresh air is drawn in from the atmosphere. The pressure in the cylinder during suction stroke is 0.9×10^5

Pa and the atmospheric pressure is $1.013 \times 10^5 \text{ N/m}^2$. The difference between the suction and atmospheric pressure is accounted for flow resistance in the suction pipe and inlet valve. Find the net work

done during the process, [7.5 Marks]

QUESTION 3

(a) A researcher asserts that he has developed a thermal engine with an efficiency of 100%. With your

knowledge of the laws of thermodynamics, give your criticisms of this engine. [4 marks]

(b) State the difference between an isolated process and an adiabatic process [2.5 marks]

(c) State the cold air standard assumptions. [2.5 marks]

(d) A group of FUPRE research students carried out an analysis of the Carnot cycle. In this analysis, these young and

~Giifiany minds gave the following information: tit the Carnot cycle operates between 600°C and 50°C, and it

produces 130kJ of work. Another group of students from Uniben determined, using the information provided by

FUPRE research students, the following: that (i) the thermal efficiency of the Carnot engine analyzed by FUPRE

students is 53.5% (ii) the heat added is 243kJ and (iii) entropy change during heat rejection process is

0.361kJ/K. Show by detailed calculations whether Uniben students are right or wrong? [8.5 marks]

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QUESTION 4

(a) What is (or are) the limitations and criticisms of the first law of thermodynamics? [2 marks]

(b) Derive an expression for the thermal efficiency of an Otto cycle [4 marks]

(c) Why is the actual efficiency of a thermodynamic cycle always less than the ideal efficiency? [2 marks]

(d) 0.5kg of air (ideal gas) executes a Carnot power cycle having a thermal efficiency of 50%. The heat transfer to

the air during the isothermal expansion is 40kJ. At the beginning of the isothermal expansion the pressure is

7bar and the volume is 0.12m³, Determine: (i) the maximum and minimum temperatures for the cycle in Kelvin. (ii)

the volume at the end of the isothermal expansion in m³. (iii) the heat transfer for each of the four processes in

kJ. For air, take $C_v = 0.718$ kJ/kgK & $C_p = 1.008$ kJ/kgK [9.5 marks]

QUESTION 5

(a) With the aid of well labeled P-v and T-s diagram, describe how pressure, volume and temperature changes within

the Dual cycle. [4.5 marks]

(b) A diesel cycle has a compression ratio of 15 and heat addition at constant pressure takes place at 6% stroke.

Find the air standard efficiency of the engine, Take ratio of specific heats to be 1.4. [5.5 marks]

(c) List and explain the processes in a Carnot cycle. [4.5 marks]

(d) Derive an expression for the efficiency of a Carnot cycle. de A \ oO

QUESTION 6 L. : yw

(a) Very seasoned Engineers had strongly chief Ye, Otto cycle and Diesel cycle, arguing that they are oversimplification of an attempt at solving complex problems, Based on this criticism, the Dual cycle was

developed. Clearly, but briefly, explain these limitations that led to the development of the Dual cycle. [5 marks]

(b) Show that the equation, $B_{\infty} = a$

force; P is pressure, ρ is density, u is velocity, ν is viscosity and x is distance. [2 marks]

(c) Compare and contrast Work and Heat? [2 marks] .

(d) From the figure below, where $V_e = 0.0048 \text{ m}^3$, $V_s = 0.0195 \text{ m}^3$, $\gamma = 1.4$ (i) what is the name of the cycle whose P - v

and T - s diagram are shown in the figure below. (ii) estimate the air standard efficiency (iii) mean effective

pressure is either dimensionally homogenous or not. Where the unit of B , is

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— [8.5 marks]

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT. OF MECHANICAL ENGINEERING i

First Semester Examination - 2025/2016 Academic Session

COURSE CODE AND TITLE: MEE 26 - BASIC THERMODYNAMICS

DATE: APRIL 19, 2026. TIME ALLOWED: 2 Hrs

"INSTRUCTIONS: -ANSWER QUESTION ONE AND ANY OTHER THREE QUESTIONS.

- STUDENTS SHOULD BE ALLOWED TO USE THEIR THERMODYNAMIC PROPERTIES TABLES.

QUESTION 1 (COMPULSORY) — thermodynamics < Wee science "therm cart.

(i) Define Thermodynamics, "e895 emd, $m =$ esel Sycleme: ersten [2 Marks]

(ii) Summarize the basic laws upon which the study of Thermodynamics is based. Eh ee, [2 Marks]

iii) - Mention the essential components required for complete definition of a thermodynamic system

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[3 Marks]

(iv) State one (1) each of the basic similarities and differences between Steady flow and Non-flow

systems. . J [2 Marks]

(v) Mention the forms in which energy exists. List transition, clear mmd work - [2 Marks]

(vii) How many processes would be contained 'in a thermodynamic cycle with three states:

Illustrate the cycle on a P - V diagram. Indicate clearly the states and processes, [4 Marks]

(viii) State the Joule's law of internal energy. (number of moles of perfect gas) [1 Mark]

(ix) List the four (4) types of equilibrium states that a thermodynamic system must achieve for it to attain a state of thermodynamic equilibrium. — [5 Marks]

(x) - A closed thermodynamic system: contains air at 60°C , and the air receives heat at constant-volume until its temperature reaches 510°C . Determine: (i) work done per kg of air (ii) change in internal energy per kg of air (iii) the amount of heat added to the system. Take $C_{v, \text{air}} = 0.718 \text{ kJ/kgK}$. [5 Marks]

QUESTION 2

(a) A piston and cylinder machine containing a fluid system has a stirring device as shown in Fig. Q2a

The piston is frictionless, and it is held down against the fluid due to atmospheric pressure of 101 kPa . The stirring device is turned at angular speed of 9500 rev/min , dissipating 1.24 kW power to the fluid. Meanwhile the piston of 0.65 m -diameter moves out 0.6 m . Determine the average torque of the stirring device against the fluid. Find also the work done due to piston movement, indicating its effect on the net work transfer for the system [5 Marks]

(vi) Is it true the assertion: "adiabatic systems are isolated"? Justify your answer: see Ks]

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Paper 2 of 2 — MEE 216 Handout

Level 200 • First Semester • Handout

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Vapour and Gas Power Cycle ①

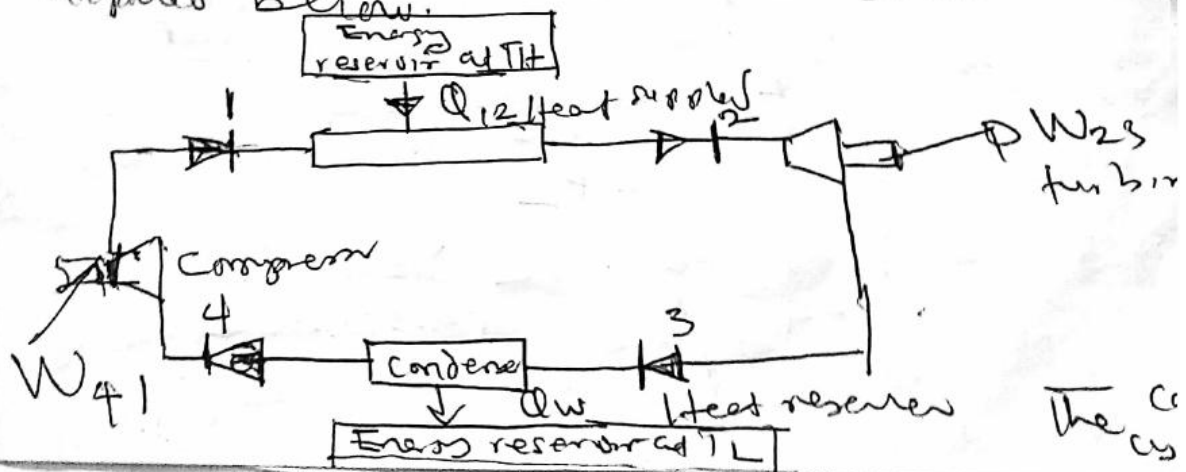
This lecture is covered with the applications of the previous laws in the thermodynamic analysis of cycles used for conversion of heat energy to mechanical work.

The Carnot cycle

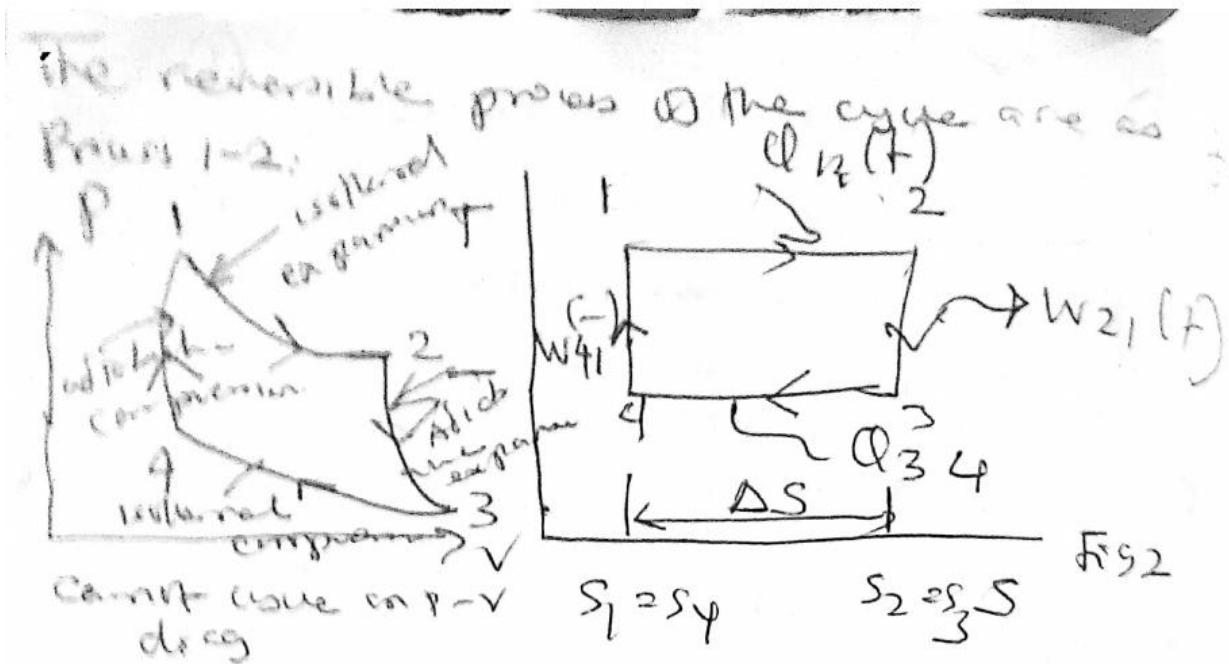
This is a cycle suggested by Sadi Carnot (1796-1837).

It is a totally reversible cycle and provides the maximum theoretical efficiency of a heat engine which cannot be achieved by actual heat engines.

The Carnot cycle consists of two reversible isothermal (constant temperature) processes joined by two reversible adiabatic (isentropic) processes. A diagrammatic sketch of a heat engine working on a Carnot cycle is depicted below.



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Carnot cycle on T-S diagram

Process 1-2: a reversible heat supply to the working fluid at constant temperature T_H and pressure P_1 .

Process 2-3: a reversible adiabatic ($S_2 = S_3$) expansion of the fluid from high temperature T_H and pressure P_1 to low temperature T_L and pressure P_2 . During this process work is done by the expanding fluid on the surroundings.

Process 3-4: A reversible heat rejection at constant temperature T_L and pressure P_2 .

Process 4-1: A reversible adiabatic (isentropic $S_4 = S_1$) compression of the fluid from the temperature T_L and pressure P_2 to its original state 1. The compressor does work on the fluid during the process.

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Referring to Fig. 2, and making use of equation
 coefficient of performance (COP) = ratio of the desired
 effect to the work input to achieve this effect

$$COP = \frac{\text{desired effect}}{\text{work input}}$$

Regardless of the working fluid employed in the cycles,
 the heat transfers are:

Heat supplied $Q_{12} = T_H \Delta S$ — (1)

Heat rejected $Q_{34} = T_L \Delta S$ — (2)

Note $\Delta S = S_2 - S_1 = S_2 - S_4$ in eqs (1) & (2)

The Carnot efficiency is given by

$$\eta_{\text{Carnot}} = \frac{\text{net work output}}{\text{heat input}} = \frac{W_{\text{net}}}{Q_{12}} \quad \text{--- (3)}$$

Application of the first law to the cycle yields

$$W_{\text{net}} = Q_{\text{net}}$$

$$W_{\text{net}} = Q_{12} - Q_{24} \quad \text{--- (4)}$$

~~Sub eqn (4) and (2) in equation~~

~~sub eq (1) and (2) in equation~~

sub eq (1) and (2) gives

$$\eta_{\text{Carnot}} = \frac{Q_{12} - Q_{24}}{Q_{12}} = 1 - \frac{Q_{24}}{Q_{12}} \quad \text{--- (5)}$$

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substituting eq. (1) and (2) in eq. (3) we have (4)

$$\eta_{\text{Carnot}} = \frac{1 - T_L \Delta S}{T_H \Delta S}$$

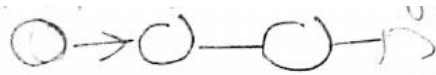
$$\boxed{\eta_{\text{Carnot}} = 1 - \frac{T_L}{T_H}} \quad \text{--- (5)}$$

From eq. (5), we can conclude that the thermal efficiency of a heat engine which operates in the Carnot cycle depends only on the temps of the 2 reservoirs b/w which heat exchanges take place. It can be observed that the higher the temp at which heat is supplied the higher the efficiency and the lower the temp at which heat is rejected the higher the efficiency.

problem

A heat engine which operates in a Carnot cycle receives 700 kJ of heat energy at 850°C. The efficiency of the heat engine is 48%. Determine the net work output, the heat rejected and the temperature at which the heat is rejected.

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solution

Heat engine operating on the Carnot cycle

$$\eta_{\text{Carnot}} = 48\% = 0.48$$

$$T_H = 850 + 273 = 1123 \text{ K}, Q_H = Q_{12} = 700 \text{ kJ}$$

= heat input

- Find (i) Net work output
 (ii) Q_L heat rejected
 (iii) T_L temperature at which heat is rejected

Solr (i) What $= ?$

$$\eta_{\text{Carnot}} = \frac{W_{\text{net}}}{Q_{12}} = 0.48 \quad \text{--- (1)}$$

$$W_{\text{net}} = 0.48 \times Q_{12} = 0.48 \times 700$$

$$W_{\text{net}} = 336 \text{ kJ}$$

(ii) $Q_L = ?$

Application of the 1st Law to the Carnot cycle gives

$$W_{\text{net}} = Q_H - Q_L$$

$$Q_L = Q_H - W_{\text{net}} = 700 - 336$$

$$Q_L = 364 \text{ kJ}$$

(iii) $T_L = ?$

$$\eta_{\text{Carnot}} = 1 - \frac{T_L}{T_H} \quad \text{--- (2)}$$

Comparing eq (1) and (2) gives

$$1 - \frac{T_L}{T_H} = 0.48$$

$$\frac{T_L}{T_H} = 1 - 0.48 = 0.52$$

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$$\frac{T_L}{T_H} = 0.52$$

$$T_L = 0.52 \times T_H = 0.52 \times 1123$$

$$T_L = 584 \text{ K or } 311^\circ\text{C}$$

Impracticability of the Carnot Steam Cycle

- ① The Carnot cycle has the max. theoretical thermal efficiency for a heat engine but the cycle is not realizable for 3 reasons
- ① All the four processes have to be reversible P517B
- ② The heat rejection and absorption have to take place with infinitesimal temp diff.
- ③ For attaining isothermal expansion, the rate of heat transfer is required to be very low

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Problems on the Carnot cycle & refrigeration

- ① A heat engine working on Carnot cycle ~~operates~~ converts one-fifth of the heat input into work. When the temp of the sink is reduced by 80%, the efficiency gets doubled. Determine the temp of source & sink.

(Ans. Temp of sink, T_2 or $T_L = 320\text{ K}$
Temp of source, $T_1 = 1.25 \times 320 = 400\text{ K}$)

- ② A reversible heat engine delivers 0.6 kW power and rejects heat energy for a reservoir at 300 K at the rate of 24 kJ/min. Calculate the engine efficiency and the temp. of the thermal reservoir supplying heat to the engine.

(Ans. Thermal efficiency, $\eta = 60\%$
 $T_1 = 750\text{ K}$)

(Ans. $\dot{Q}_1 = 215.9736\text{ kJ/s}$)

- ③ An engine mounted on a ship has a thermal efficiency of 80% of that of the corresponding Carnot cycle. The engine receives heat from the sea at 300 K and rejects heat to the atmosphere at 280 K. The work output from the engine is dissipated through an agitator to heat 500 kg of sea water to 355 K. What quantity of heat must be extracted from the sea water to provide the required heating effect. Take specific heat of sea water $c_p = 4.186\text{ kJ/kg K}$.

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Example on the steam tables

Tables of Thermodynamics Properties
The relationship between pressure, volume and temperature and other thermodynamic properties of several pure substances are usually presented in tabular or graphical forms.

Examples

1. Use the steam tables to determine the following:
 - a) the specific internal energy of steam of 90% quality at a pressure of 4.0 bar
 - b) the dryness fraction of steam at 1 bar with a specific enthalpy of 2340 kJ/kg.

Solution

working fluid, steam, $x = 0.90$, $P = 4.0 \text{ bar}$
 $u = ?$ (to find specific internal energy)

$$u = u_f + x(u_g - u_f)$$

The 90% quality indicates that the fluid is in the wet vapour region, hence,

$$u = u_f + x(u_g - u_f)$$

The saturated water/steam table p.54 is consulted to obtain u_f and u_g at P.

Hence $u = u_f + x(u_g - u_f)$

$$605 + 0.90(2554 - 605)$$

$$u = 2359 \text{ kJ/kg}$$

3552 kJ/kg heat

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⑤ Working fluid: steam at $P_2 = 1 \text{ bar}$, $h = 2340 \text{ kJ/kg}$ 011

Find, $x =$ dryness fraction or quality
By requesting for the dryness fraction or quality means that the fluid is in the wet vapour region, so that

$$h = h_f + x h_{fg}$$

from which

$$x = \frac{h - h_f}{h_{fg}}$$

Look at the saturated/water steam table
(p. 44)

at $P_2 = 1 \text{ bar}$, $h_f = 417 \text{ kJ/kg}$

$$h_{fg} = 2258 \text{ kJ/kg} = (h_g - h_f)$$

$$x = \frac{2340 - 417}{2258} = 0.85$$

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Saturated water and Steam, specific enthalpy

P (bar)	P (MPa)	v_g (m ³ /kg)	u_f kJ/kg	u_g kJ/kg	h_f	h_g	s_f	s_g	s_g
1.0			417	417					
4.0			605	2284					

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